

WEST BENGAL JEE 2026

PHYSICS MOCK TEST QUESTION

General Instructions

- This paper contains 75 questions divided into three categories.
- **Category-I:** 1 Mark each. Only one option correct. Negative marking: -0.25.
- **Category-II:** 2 Marks each. Only one option correct. Negative marking: -0.5.
- **Category-III:** 2 Marks each. One or more options may be correct. No negative marking.
- Partial marks are awarded for partially correct answers in Category-III.

Category: I (Q: 1 to 50)

Only one option is correct. Correct response yields 1 mark. Incorrect response yields -0.25 marks.

1. The dimensional formula for Planck's constant is identical to that of:

- (A) Linear momentum
- (B) Angular momentum
- (C) Energy
- (D) Power

Ans: (B)

2. A block of mass m is placed on a smooth inclined plane of inclination θ . The inclined plane is given an acceleration a horizontally so that the block does not slide down. The value of a is:

- (A) $g \tan\theta$
- (B) $g \sin\theta$
- (C) $g \cos\theta$
- (D) $g \cot\theta$

Ans: (A)

3. In a thermodynamic process, the pressure of a fixed mass of gas is changed such that the gas releases 20 J of heat and 8 J of work is done on the gas. If the initial internal energy was 30 J, the final internal energy is:

- (A) 18 J
- (B) 42 J
- (C) 58 J
- (D) 2 J

Ans: (A)

4. Two point charges $+q$ and $-q$ are placed at a distance $2l$ apart. The electric potential at a point on the equatorial line at a distance r from the center of the dipole is:

- (A) Zero
- (B) $p / (4\pi\epsilon_0 r^2)$
- (C) $p / (4\pi\epsilon_0 r^3)$
- (D) $2p / (4\pi\epsilon_0 r^2)$

Ans: (A)

5. A ray of light passes through an equilateral glass prism such that the angle of incidence equals the angle of emergence. If the angle of emergence is $3/4$ times the angle of the prism, the refractive index of the glass is:

- (A) $\sqrt{2}$
- (B) $\sqrt{3}$
- (C) 1.5
- (D) 1.33

Ans: (A)

6. The half-life of a radioactive substance is 20 minutes. The time taken between 20% and 80% decay will be:

- (A) 20 minutes
- (B) 30 minutes
- (C) 40 minutes
- (D) 60 minutes

Ans: (C)

7. The ratio of the radii of gyration of a circular disc about a tangential axis in its plane and of a circular ring of the same radius about a tangential axis in its plane is:

- (A) $\sqrt{5} : \sqrt{6}$
- (B) $2 : 3$
- (C) $\sqrt{2} : \sqrt{3}$
- (D) $1 : \sqrt{2}$

Ans: (A)

8. An electromagnetic wave is propagating along the +z axis. The direction of the electric field (E) and magnetic field (B) can be represented by:

- (A) E along +x, B along +y
- (B) E along +y, B along +x
- (C) E along +z, B along +x
- (D) E along +x, B along +z

Ans: (A)

9. A particle is projected with a velocity u making an angle θ with the horizontal. The magnitude of its velocity at the highest point of its trajectory is:

- (A) $u \sin\theta$
- (B) $u \cos\theta$
- (C) u
- (D) Zero

Ans: (B)

10. A car of mass m is moving on an unbanked circular track of radius R. If the coefficient of static friction is μ , the maximum safe speed is:

- (A) $\sqrt{\mu R g}$
- (B) $\mu R g$
- (C) $\sqrt{\mu g / R}$
- (D) $\mu g / R$

Ans: (A)

11. A spring of force constant k is cut into two equal halves. The force constant of each half is:

(A) $k/2$

(B) k

(C) $2k$

(D) $4k$

Ans: (C)

12. The escape velocity from the Earth is v_e . If a planet has twice the radius and twice the mean density of Earth, its escape velocity will be:

(A) v_e

(B) $2 v_e$

(C) $2\sqrt{2} v_e$

(D) $4 v_e$

Ans: (C)

13. The ratio of specific heats $\gamma = C_p/C_v$ for a diatomic gas at normal temperature is:

(A) 1.67

(B) 1.40

(C) 1.33

(D) 1.28

Ans: (B)

14. Water rises in a vertical capillary tube up to a height h . If the tube is inclined at an angle of 45° with the vertical, the length of the water column will be:

(A) h

(B) $h / \sqrt{2}$

(C) $h \sqrt{2}$

(D) $2h$

Ans: (C)

15. The magnetic flux linked with a coil varies with time t as $\phi = 3t^2 + 4t + 9$ Wb. The magnitude of the induced emf at $t = 2$ s is:

- (A) 8 V
- (B) 16 V
- (C) 22 V
- (D) 32 V

Ans: (B)

16. An ideal gas undergoes an isothermal expansion at temperature T . The work done by n moles of the gas in expanding from V_1 to V_2 is:

- (A) $nRT \ln(V_2/V_1)$
- (B) $nRT \ln(V_1/V_2)$
- (C) $nRT(V_2 - V_1)$
- (D) $(nR/T) \ln(V_2/V_1)$

Ans: (A)

17. The efficiency of a Carnot engine working between temperatures of 127°C and 27°C is:

- (A) 25%
- (B) 50%
- (C) 75%
- (D) 100%

Ans: (A)

18. The distance between two consecutive nodes in a stationary wave is:

- (A) λ
- (B) $\lambda/2$
- (C) $\lambda/4$
- (D) 2λ

Ans: (B)

19. According to Stokes' law, the terminal velocity of a small spherical body of radius r falling through a viscous medium is proportional to:

- (A) r
- (B) r^2
- (C) r^3
- (D) $1/r$

Ans: (B)

20. The dimensions of Young's modulus of elasticity are the same as those of:

- (A) Force
- (B) Pressure
- (C) Strain
- (D) Work

Ans: (B)

21. The moment of inertia of a solid sphere of mass M and radius R about its diameter is:

- (A) $(2/5) MR^2$
- (B) $(2/3) MR^2$
- (C) $(1/2) MR^2$
- (D) MR^2

Ans: (A)

22. According to Kepler's third law, the square of the time period of revolution of a planet is proportional to the:

- (A) semi-major axis
- (B) square of the semi-major axis
- (C) cube of the semi-major axis
- (D) inverse square of the semi-major axis

Ans: (C)

23. The electric field outside an infinite thin uniformly charged planar sheet with surface charge density σ is:

- (A) σ / ϵ_0
- (B) $\sigma / (2\epsilon_0)$
- (C) $2\sigma / \epsilon_0$
- (D) Zero

Ans: (B)

24. The equivalent capacitance of two capacitors C_1 and C_2 connected in series is:

- (A) $C_1 + C_2$
- (B) $(C_1 C_2) / (C_1 + C_2)$
- (C) $(C_1 + C_2) / (C_1 C_2)$
- (D) $|C_1 - C_2|$

Ans: (B)

25. The drift velocity of free electrons is v_d when current I flows. If both the radius and the current are doubled, the drift velocity will be:

- (A) $v_d / 4$
- (B) $v_d / 2$
- (C) v_d
- (D) $2 v_d$

Ans: (B)

26. Kirchhoff's voltage law (KVL) is a consequence of the law of conservation of:

- (A) Charge
- (B) Momentum
- (C) Energy
- (D) Angular momentum

Ans: (C)

27. The magnetic field B at a distance r from a long straight wire carrying current I is proportional to:

- (A) $I \cdot r$
- (B) I / r^2
- (C) I^2 / r
- (D) I / r

Ans: (D)

28. The force experienced by a charge q moving with velocity v in a uniform magnetic field B is:

- (A) $q(v \cdot B)$
- (B) $q(v \times B)$
- (C) $q(B \times v)$
- (D) $(v \times B) / q$

Ans: (B)

29. The resonant frequency f_r of a series LCR circuit is:

- (A) $1 / (2\pi\sqrt{LC})$
- (B) $1 / \sqrt{LC}$
- (C) $2\pi\sqrt{LC}$
- (D) $(1/2\pi) \sqrt{L/C}$

Ans: (A)

30. The core of a transformer is laminated to reduce energy loss due to:

- (A) Eddy currents
- (B) Hysteresis
- (C) Flux leakage
- (D) Copper loss

Ans: (A)

31. Which of the following electromagnetic waves has the longest wavelength?

- (A) X-rays
- (B) Ultraviolet rays
- (C) Microwaves
- (D) Radio waves

Ans: (D)

32. A convex mirror always produces an image that is:

- (A) Real, inverted, and magnified
- (B) Virtual, erect, and magnified
- (C) Virtual, erect, and diminished
- (D) Real, erect, and diminished

Ans: (C)

33. In Young's Double Slit Experiment, the fringe width β is given by:

- (A) $\lambda d / D$
- (B) $\lambda D / d$
- (C) $D / (\lambda d)$
- (D) $d / (\lambda D)$

Ans: (B)

34. According to Brewster's law, the refractive index μ of a medium is related to the polarizing angle i_p by:

- (A) $\mu = \sin(i_p)$
- (B) $\mu = \cos(i_p)$
- (C) $\mu = \tan(i_p)$
- (D) $\mu = \cot(i_p)$

Ans: (C)

35. The energy E of a photon of momentum p is given by (c = speed of light):

- (A) $E = pc$
- (B) $E = p/c$
- (C) $E = pc^2$
- (D) $E = p^2/(2m)$

Ans: (A)

36. The de Broglie wavelength of an electron accelerated through V volts is approximately:

(A) $12.27 / \sqrt{V} \text{ \AA}$

(B) $1.227 / \sqrt{V} \text{ \AA}$

(C) $12.27 \sqrt{V} \text{ \AA}$

(D) $12.27 / V \text{ \AA}$

Ans: (A)

37. In the Bohr model of the hydrogen atom, the radius of the nth orbit is proportional to:

(A) n

(B) n^2

(C) $1/n$

(D) $1/n^2$

Ans: (B)

38. The half-life $T_{1/2}$ of a radioactive substance is related to its decay constant λ by:

(A) $T_{1/2} = \lambda \ln(2)$

(B) $T_{1/2} = \ln(2) / \lambda$

(C) $T_{1/2} = \lambda / \ln(2)$

(D) $T_{1/2} = 1 / \lambda$

Ans: (B)

39. The maximum binding energy per nucleon occurs for the nucleus of:

(A) Helium (^4He)

(B) Oxygen (^{16}O)

(C) Iron (^{56}Fe)

(D) Uranium (^{238}U)

Ans: (C)

40. In a reverse-biased p-n junction, the width of the depletion layer:

(A) Decreases

(B) Increases

(C) Remains the same

(D) First increases then decreases

Ans: (B)

41. The logic gate that produces a high (1) output only when all its inputs are high (1) is:

- (A) OR gate
- (B) AND gate
- (C) NOR gate
- (D) NAND gate

Ans: (B)

42. A Zener diode is primarily used as:

- (A) An amplifier
- (B) A voltage regulator
- (C) A rectifier
- (D) An oscillator

Ans: (B)

43. If the percentage error in measuring the radius of a sphere is 2%, the percentage error in its volume is:

- (A) 2%
- (B) 4%
- (C) 6%
- (D) 8%

Ans: (C)

44. The magnitude of the cross product of two parallel vectors is:

- (A) Product of their magnitudes
- (B) Zero
- (C) One
- (D) Infinity

Ans: (B)

45. The angle of repose on a rough inclined plane is equal to the:

- (A) Angle of incidence
- (B) Angle of friction
- (C) Critical angle
- (D) Angle of deviation

Ans: (B)

46. When a gas undergoes an adiabatic expansion, its temperature:

- (A) Increases
- (B) Decreases
- (C) Remains constant
- (D) Depends on the specific heat

Ans: (B)

47. The SI unit of magnetic dipole moment is:

- (A) A m
- (B) A m²
- (C) T m A⁻¹
- (D) Wb m²

Ans: (B)

48. A particle performs SHM with amplitude A. Its kinetic energy equals its potential energy when its displacement from the mean position is:

- (A) A/2
- (B) A/√2
- (C) A
- (D) Zero

Ans: (B)

49. The power factor of a purely resistive AC circuit is:

- (A) Zero
- (B) 0.5
- (C) 1
- (D) Infinity

Ans: (C)

50. The core of an electromagnet is made of soft iron because soft iron has:

- (A) High retentivity and high coercivity
- (B) Low retentivity and low coercivity
- (C) High retentivity and low coercivity
- (D) Low retentivity and high coercivity

Ans: (B)

Category: II (Q: 51 to 65)

Only one option is correct. Correct response yields 2 marks. Incorrect response yields -0.5 marks.

51. A solid sphere rolls down two different inclined planes of the same height but different angles of inclination. In both cases, the sphere:

- (A) reaches the bottom with the same speed and in the same time.
- (B) reaches the bottom with the same speed but different times.
- (C) reaches the bottom with different speeds but the same time.
- (D) reaches the bottom with different speeds and different times.

Ans: (B)

52. A simple pendulum has a time period T_1 on the earth's surface and T_2 at a height R above the earth's surface, where R is earth's radius. The ratio T_2/T_1 is:

- (A) 1
- (B) $\sqrt{2}$
- (C) 2
- (D) 4

Ans: (C)

53. A parallel plate capacitor is charged by a battery which is then disconnected. A dielectric slab is now inserted between the plates. Which quantity remains constant?

- (A) Capacitance
- (B) Potential difference
- (C) Electric field
- (D) Charge

Ans: (D)

54. A particle of mass m and charge q is projected with velocity v perpendicular to a uniform magnetic field B . The work done by the magnetic force after one revolution is:

- (A) mv^2/r
- (B) $qvB(2\pi r)$
- (C) Zero
- (D) $(1/2)mv^2$

Ans: (C)

55. In a Young's double-slit experiment, the intensity of light where the path difference is λ is K. The intensity where the path difference is $\lambda/4$ will be:

- (A) K
- (B) $K/2$
- (C) $K/4$
- (D) Zero

Ans: (B)

56. A block of mass m is pulled along a horizontal frictionless surface by a thick rope of mass $m/2$. If a force F is applied at the free end of the rope, the force exerted by the rope on the block is:

- (A) F
- (B) $2F/3$
- (C) $F/3$
- (D) $3F/2$

Ans: (B)

57. Two spheres of the same material have radii R and $2R$. If they are heated to the same temperature and left in identical surroundings, the ratio of their initial rates of cooling is:

- (A) 1:1
- (B) 2:1
- (C) 1:2
- (D) 4:1

Ans: (B)

58. An ideal gas heat engine operates in a Carnot cycle between 227°C and 127°C . It absorbs 600 kcal at the higher temperature. The amount of heat converted into work is:

- (A) 120 kcal
- (B) 240 kcal
- (C) 480 kcal
- (D) 600 kcal

Ans: (A)

59. The equivalent resistance between diagonally opposite corners of a wire cube of resistance R on each edge is:

- (A) $5R/6$
- (B) $7R/12$
- (C) $3R/4$
- (D) R

Ans: (A)

60. The magnetic susceptibility of a paramagnetic substance at -73°C is 0.0060 . Its susceptibility at -173°C will be:

- (A) 0.0030
- (B) 0.0120
- (C) 0.0180
- (D) 0.0045

Ans: (B)

61. In a series LCR circuit connected to an AC source, the voltage across R , L , and C is 10 V each. If the capacitor is short-circuited, the new voltage across the inductor will be:

- (A) 10 V
- (B) $10/\sqrt{2}\text{ V}$
- (C) $10/3\text{ V}$
- (D) 0

Ans: (B)

62. If the momentum of an electron is changed by p , the de Broglie wavelength associated with it changes by 0.5% . The initial momentum of the electron is:

- (A) 100 p
- (B) 200 p
- (C) 400 p
- (D) $p/200$

Ans: (B)

63. A radioactive sample with half-life T contains N_0 active nuclei at $t=0$. The number of nuclei decayed in time $t = T/2$ is:

- (A) $N_0 / \sqrt{2}$
- (B) $N_0 (1 - 1/\sqrt{2})$
- (C) $N_0 / 2$
- (D) $N_0 / 4$

Ans: (B)

64. Two thin lenses of focal lengths f_1 and f_2 are in contact and coaxial. The power of the combination is:

- (A) $f_1 + f_2$
- (B) $(f_1 f_2) / (f_1 + f_2)$
- (C) $1/f_1 + 1/f_2$
- (D) f_1 / f_2

Ans: (C)

65. The minimum deviation produced by a hollow prism filled with a liquid is 30° . If the refracting angle of the prism is 60° , the refractive index of the liquid is:

- (A) 1.5
- (B) 1.33
- (C) 1.414
- (D) 1.61

Ans: (C)

Category: III (Q: 66 to 75)

One or more options may be correct. Correct response yields 2 marks. No negative marking.

66. A satellite is orbiting the earth in a circular orbit. If its mass is suddenly doubled without changing its orbital speed or radius, which statements are correct?

- (A) Its time period will remain unchanged.
- (B) Its kinetic energy will be doubled.
- (C) Its potential energy will remain unchanged.
- (D) Its angular momentum will be doubled.

Ans: (A, B, D)

67. Which of the following equations represent a traveling wave?

- (A) $y = A \sin(kx - \omega t)$
- (B) $y = A \cos(kx + \omega t)$
- (C) $y = A \sin(kx) \cos(\omega t)$
- (D) $y = A e^{-(x - vt)^2}$

Ans: (A, B, D)

68. For a photoelectric effect experiment, which of the following statements are true?

- (A) The stopping potential depends on the intensity of incident light.
- (B) The stopping potential depends on the frequency of incident light.
- (C) The saturation current depends on the intensity of incident light.
- (D) The photoelectric emission is an instantaneous process.

Ans: (B, C, D)

69. In an LCR series circuit connected to an AC source at resonance:

- (A) The impedance of the circuit is minimum.
- (B) The current in the circuit is maximum.
- (C) The voltage across the inductor and capacitor are equal and opposite in phase.
- (D) The power factor of the circuit is zero.

Ans: (A, B, C)

70. Let V and E denote the gravitational potential and field at a point, respectively. It is possible that:

- (A) $V = 0$ and $E = 0$
- (B) $V = 0$ and $E \neq 0$
- (C) $V \neq 0$ and $E = 0$
- (D) $V \neq 0$ and $E \neq 0$

Ans: (A, B, C, D)

71. A particle moves in a circular path with continuously increasing speed. Which of the following are correct?

- (A) Its acceleration vector is strictly perpendicular to its velocity vector.
- (B) Its acceleration vector is not perpendicular to its velocity vector.
- (C) The work done by the net force is positive.
- (D) Its angular momentum about the center is conserved.

Ans: (B, C)

72. In an isolated charged capacitor with a dielectric:

- (A) If the dielectric is removed, the potential difference increases.
- (B) If the dielectric is removed, the energy stored increases.
- (C) If the dielectric is removed, the charge increases.
- (D) If the dielectric is removed, the electric field decreases.

Ans: (A, B)

73. Which of the following quantities are quantized in Bohr's model of a hydrogen atom?

- (A) Linear momentum
- (B) Angular momentum
- (C) Kinetic energy
- (D) Total energy

Ans: (B, C, D)

74. A ray of light is incident from a denser medium to a rarer medium. The critical angle is θ_c .

- (A) If angle of incidence $i > \theta_c$, total internal reflection occurs.
- (B) If $i < \theta_c$, the ray is partially reflected and partially refracted.
- (C) The refractive index of denser medium with respect to rarer medium is $1/\sin(\theta_c)$.
- (D) Total internal reflection can occur when light travels from rarer to denser medium.

Ans: (A, B, C)

75. In the decay process: $A \rightarrow B + \alpha$, $B \rightarrow C + \beta^-$:

- (A) Mass number of A is 4 more than that of C.
- (B) Atomic number of A is 1 more than that of C.
- (C) A and C are isotopes.
- (D) B and C are isobars.

Ans: (A, B, D)